

AI-Powered Image Generation System using AI

Vidya Jamadade¹, Madubala Ghodake², Samriddhi Katakdhond³, Prof. Vaibhav Godase⁴

^{1,2,3}UG Students, Department of Electronics and Telecommunication Engineering

⁴Assistant Professor, Department of Electronics and Telecommunication Engineering

SKN Sinhgad College of Engineering, Pandharpur

jamadadevidya2601@gmail.com, ghodakemadhubalal@gmail.com,

ksamriddhi05@gmail.com, vaibhav.godase@sknscoe.ac.in

Abstract: *Advancements in artificial intelligence have opened new pathways for creating visual content through language. This project presents an intelligent image generation system that converts textual descriptions into lifelike and high-quality images using AI state-of-the-art generative models. By transforming simple text prompts into detailed and contextually relevant visuals, the system empowers users to bring their creative ideas to life without requiring prior design expertise.*

Employing deep learning and natural language processing (NLP) techniques, the system accurately interprets linguistic inputs, identifies semantic relationships, and translates descriptive text into meaningful visual elements. Its intuitive and responsive interface ensures a seamless user experience, making AI-driven image generation useful across a wide range of fields such as digital design, marketing, education, entertainment, and research.

In addition, the project investigates key issues related to AI-based creativity, including prompt accuracy, model bias, output reliability, and image quality enhancement. It also explores ethical aspects and future prospects of AI in the creative sector, showcasing how AI generative technology can revolutionize digital artistry and strengthen collaboration between humans and intelligent systems.

Keywords: Stability AI, Text-to-image synthesis, Prompt-based image generation

I. INTRODUCTION

Artificial Intelligence (AI) has emerged as one of the most transformative technologies of the 21st century, profoundly influencing industries across science, business, and the arts. From automating complex data processes to enabling intelligent decision-making and content creation, AI continues to redefine how humans interact with technology. One of the most remarkable advancements within this domain is AI-driven image generation, a field that combines natural language understanding with visual creativity to produce high-quality, realistic images directly from textual descriptions. AI-powered image generation bridges the gap between linguistic expression and digital visualization. By simply describing an idea or scene in words, users can generate detailed and visually compelling images that accurately represent their imagination. This capability has opened new possibilities for individuals and organizations alike—empowering non-designers to create professional-quality visuals and enabling artists, marketers, and educators to experiment with new forms of creative expression.

The AI-Powered Image Generation System using AI is designed to leverage AI advanced generative models to achieve this goal. The system employs cutting-edge deep learning and natural language processing (NLP) techniques to interpret text prompts, analyse contextual meaning, and translate linguistic patterns into coherent visual representations. By mapping the semantic relationships between words and visual features, the system ensures that generated images align closely with user intent and maintain high aesthetic and contextual accuracy. Moreover, the project addresses several key challenges inherent to AI-based generative systems, including prompt interpretation, model bias, consistency of output, and optimization of image quality. Ethical considerations are also explored, particularly those concerning intellectual property, authenticity of AI-generated art, and the responsible use of generative models.



II. LITERATURE SURVEY

2.1 Overview and scope:

Text-to-image generation has matured from early GAN-based experiments into a robust subfield of multimodal generative AI. Modern systems combine large-scale image-text of conditional synthesis methods to convert natural-language prompts into high-fidelity images. AI lineage (CLIP for crossmodal representation and DALL-E variants for synthesis) exemplifies this shift: models now focus both on semantic faithfulness to prompts and on photorealism or stylistic control depending on the task. Recent survey papers emphasize diffusion models and transformer-based priors as the dominant paradigms for achieving state-of-the-art results.

2.2 Datasets and feature engineering:

Large, noisy web-scale image-text corpora (e.g., LAION) and curated benchmarks (MS-COCO, Conceptual Captions) remain the backbone of contemporary research. Open datasets such as LAION400M / LAION-5B provide billions of image-text pairs used to train multimodal encoders and generative decoders, but their uncured nature raises content, copyright, and safety issues that researchers must address during dataset construction and filtering. Several recent incidents and follow-ups highlight the need for stricter content filtering and provenance checks when using web-scraped corpora.

2.3 Methodological trends and algorithms

The AI-powered image generation system uses advanced generative models to convert text prompts into high-quality images. Diffusion models provide stable training and fine-grained control for text-conditioned image synthesis, while transformer priors with two-stage decoders map text embedding into image latent before pixel generation, as seen in DALL-E-2. GANs and hybrid approaches are occasionally used for style transfer or real-time tasks. The system also integrates safety and evaluation modules, including prompt sanitization, NSFW filtering, and copyright checks, ensuring both ethical and high-fidelity outputs. This combined approach enables accurate, creative, and responsible image generation from textual descriptions.

2.4 Fertilizer recommendation approaches:

Nutrient status estimation and Prescriptive recommendation. Nutrient status estimation predicts crop nutrient demand or soil nutrient levels (N, P, K) using sensor data, spectral indices, or soil tests. Prescriptive recommendation suggests the type and quantity of fertilizer based on target yield or soil health, often employing regression models, classification of nutrient sufficiency, or optimization techniques. While these methods can provide accurate site-specific recommendations, integrating long-term soil health and environmental impacts remains a challenge.

2.5 Integration with decision-support, GIS, and IoT systems:

Modern fertilizer recommendations and crop management systems increasingly integrate cloud-based analytics, GIS, and IoT technologies to deliver timely and precise guidance to farmers. IoT sensors measure soil moisture, nutrient levels, and environmental conditions in real time, while GIS and remote sensing provide spatial context for large-scale or within-field variability. Combining these data streams allows for site-specific recommendations, improved prediction accuracy, and operational efficiency. Despite challenges such as sensor maintenance, connectivity issues, and user adoption, these integrated systems enhance the robustness and scalability of agricultural decision support.

2.6 Measures of evaluation and methods of validation

In fertilizer recommendation and crop prediction systems, performance is typically assessed using quantitative metrics such as RMSE, MAE, and R^2 for continuous variables like yield or nutrient levels, and accuracy, precision, recall, or F1-score for classification tasks such as nutrient sufficiency categories. Validation methods commonly include cross-validation, multi-year holdout testing, and train-test splits to ensure reliability and reduce overfitting. However, many studies rely on limited temporal or geographic data, which can inflate performance metrics. Ensuring transferability across different agroclimatic zones remains a critical challenge for robust model deployment.



2.7 Difficulties and gaps

Despite advances in fertilizer recommendation and crop prediction systems, several challenges remain. Data limitations, such as small-scale experiments and a lack of standardized, long-term datasets, restrict model generalization. Transferability issues arise because models developed in one region often perform poorly in another due to soil and management variability. Explainability and agronomic relevance are critical, as black-box models must produce interpretable recommendations to gain farmer trust. Additionally, most systems focus on short-term yield or profit, rarely optimizing for long-term soil health or environmental impact. Operational challenges, including sensor costs, connectivity, and user adoption, also hinder widespread deployment.

2.8 Synopsis

The literature indicates that machine learning and deep learning techniques can significantly improve crop prediction and fertilizer recommendation when high-quality, multimodal data are available. Classical ML methods like Random Forests and Gradient Boosting provide interpretable and robust predictions, while deep learning models, including CNNs and LSTMs, effectively handle spatiotemporal and remote-sensing data. However, challenges such as limited standardized datasets, lack of cross-region validation, poor explainability, and insufficient integration of long-term soil health and environmental objectives remain. Addressing these gaps—through better data collection, model transferability, and practical deployment strategies—will be essential for reliable, scalable, and sustainable agricultural decision-support systems.

III. METHODOLOGY

FIG.1 BLOCK DIAGRAM OF TEXT TO IMAGE GENERATION

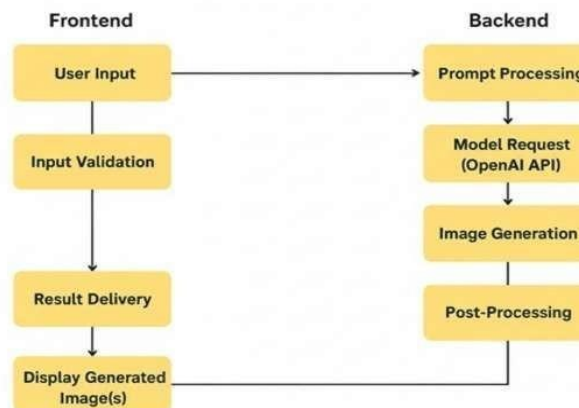


FIG.1: BLOCK DIAGRAM

Frontend:

User Input: This is where the user interacts with the system, providing text prompts or other specifications for the image they want to generate. It's the initial point of contact for the user.

Input Validation: Before processing, the system checks the user's input to ensure it meets certain criteria, such as character limits, appropriate language, or valid formats. This step helps prevent errors and ensures the prompt can be effectively used by the backend.

Result Delivery: Once the image is generated by the backend, this block handles sending the final image data back to the user's interface. It prepares the image for display.

Display Generated Image(s): This is the final step on the user's side, where the generated image is rendered and shown to the user on their screen or device.



Backend:

Prompt Processing: The user's input prompt is analysed and potentially refined or transformed into a format best understood by the image generation model. This might involve tokenization or embedding the text.

Model Request (Stability AI API): This block sends the processed prompt to the AI API (or a similar large language model API) to request the generation of an image based on the given description. It acts as the bridge to the core AI model.

Image Generation: This is the core AI process where the model interprets the prompt and creates the actual visual content. It leverages deep learning algorithms to synthesize an image from the textual description.

Post-Processing: After the image is generated, this step might involve enhancing the image, applying filters, resizing, or optimizing it for delivery and display. It ensures the generated image is ready for the user.

IV. RESULT

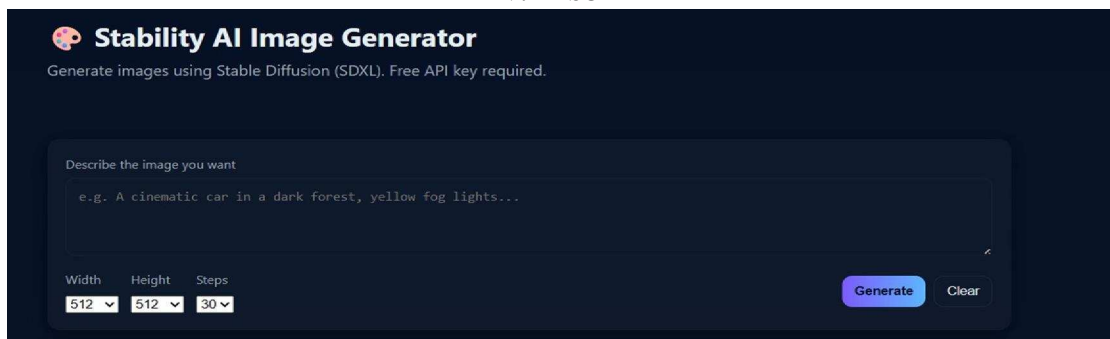


FIG.2 IMAGE GENERATION INPUT WINDOW



FIG. 3 TEXT TO IMAGE GENERATION OUTPUT WINDOW

V. DISCUSSIONS

The proposed AI-powered image generation system demonstrates the practical potential of combining deep learning and natural language processing to create images directly from textual prompts. The model efficiently interprets user input, capturing both semantic meaning and stylistic cues, which allows it to produce visually coherent and contextually relevant images. This highlights the capability of modern AI systems to serve as collaborative creative tools, augmenting human imagination rather than replacing it.

The system's usability and accessibility are significant advantages. Users without design expertise can generate high-quality visuals in real time, making it valuable for applications in marketing, education, digital media, and concept visualization. The integration of prompt-based guidance with AI models enables iterative refinement, giving users control over output style, composition, and content details.

Such interactivity enhances the creative workflow and reduces dependency on manual design processes



However, several challenges remain. Model performance is highly dependent on the diversity and quality of the training dataset, which can limit performance for unusual or highly specific prompts. Generated images may also reflect biases present in the training data or fail to meet ethical standards if content moderation is not applied. Additionally, extreme or highly abstract prompts may yield outputs that are partially incoherent or lack the intended detail.

Future improvements could include expanding the training dataset with more varied and culturally inclusive images, developing hybrid architectures to enhance fidelity and creativity, and implementing adaptive user feedback mechanisms to refine outputs over time. Overall, the project illustrates how AI-driven image generation can transform digital content creation, making creative processes faster, more accessible, and highly scalable.

VI. CONCLUSION

The AI-Powered Image Generation System using AI demonstrates the significant impact of artificial intelligence on creative processes, enabling the generation of high-quality, realistic, and contextually accurate images directly from textual prompts. By leveraging advanced deep learning architectures and natural language processing techniques, the system effectively translates human ideas into visual content, making creative workflows more efficient, interactive, and accessible to users without specialized design skills. The project highlights the versatility of AI in various domains, including digital marketing, education, entertainment, graphic design, and research, where rapid visualization of concepts can enhance productivity and innovation. The interactive nature of the platform allows users to iteratively refine outputs, explore multiple design variations, and customize style and composition, further bridging the gap between imagination and execution.

Despite these advancements, challenges such as dependency on training data quality, handling abstract or complex prompts, and mitigating biases in generated images remain areas for improvement. Ethical considerations, content moderation, and copyright concerns must also be addressed to ensure responsible deployment. Future directions include integrating larger and more diverse datasets, employing hybrid architectures that combine diffusion models, transformers, and GANs, and incorporating adaptive feedback mechanisms to improve fidelity and user control. With these enhancements, AI-powered image generation can become an even more reliable, creative, and user-friendly tool. Overall, this project underscores the transformative potential of AI in augmenting human creativity, accelerating content creation, and enabling novel applications across multiple industries, ultimately marking a shift toward more intelligent and automated digital artistry.

REFERENCES

- [1] Chand, C., & Prashant (2025). AI-Powered Text-to-Image Generation Using Stable Diffusion and Flask. International Journal of Innovative Science and Research Technology.
- [2] 'Patel, J., Tale, K., Bidwe, R.V., Mishra, S., Deshmukh, G., & Shinde, S.S. (2024). PromptArt: AI-Powered Image Generation. 2024 8th International Conference on Computing, Communication, Control and Automation (ICCUBEA), 1-6.
- [3] S A, P. (2025). Dynamic Visual Creation: Implementing Text to Image Generation. INTERNATIONAL JOURNAL OF SCIENTIFIC RESEARCH IN ENGINEERING AND MANAGEMENT.
- [4] 'Neshaei, S.P., Rietsche, R., Su, X., & Wambsganss, T. (2024). Enhancing Peer Review with AI-Powered Suggestion Generation Assistance: Investigating the Design Dynamics. Proceedings of the 29th International Conference on Intelligent User Interfaces.
- [5] 'Zhao, K., Nguyen, M., & Yan, W.Q. (2024). Evaluating Accuracy and Efficiency of Fruit Image Generation Using Generative AI Diffusion Models for Agricultural Robotics. 2024 39th International Conference on Image and Vision Computing New Zealand (IVCNZ), 1-6.
- [6] 'Banjade, S., Patel, H., & Pokhrel, S. (2024). Empowering Education by Developing and Evaluating a Generative AI-Powered Tutoring System for Enhanced Student Learning. Journal of Artificial Intelligence and Capsule Networks.



- [7] Azuaje, G., Liew, K., Buening, R., She, W.J., Siriaraya, P., Wakamiya, S., & Aramaki, E. (2023). Exploring the use of AI text-to-image generation to downregulate negative emotions in an expressive writing application. *Royal Society Open Science*, 10.
- [8] Wang, Z., Huang, Y., Song, D., Ma, L., & Zhang, T. (2024). Prompt Charm: Text-to-Image Generation through Multi-modal Prompting and Refinement. *Proceedings of the 2024 CHI Conference on Human Factors in Computing Systems*.
- [9] Brisco, R., Hay, L., & Dhami, S.K. (2023). EXPLORING THE ROLE OF TEXT-TOIMAGE AI IN CONCEPT GENERATION. *Proceedings of the Design Society*, 3, 1835 - 1844.
- [10] Na, S., Do, M., Yu, K., & Kim, J. (2022). Realistic Image Generation from Text by Using BERT-Based Embedding. *Electronics*.
- [11] Hao, Y., Chi, Z., Dong, L., & Wei, F. (2022). Optimizing Prompts for Text-to-Image Generation. *ArXiv*, abs/2212.09611.
- [12] Wu, S., Sun, M., Wang, W., Wang, Y., & Liu, J. (2025). VisualPrompter: Prompt Optimization with Visual Feedback for Text-to-Image Synthesis. *ArXiv*, abs/2506.23138.
- [13] Hei, N., Guo, Q., Wang, Z., Wang, Y., Wang, H., & Zhang, W. (2024). A User-Friendly Framework for Generating Model-Preferred Prompts in Text-to-Image Synthesis. *ArXiv*, abs/2402.12760.
- [14] Brade, S., Wang, B., Sousa, M., Oore, S., & Grossman, T. (2023). Promptify: Text-to-Image Generation through Interactive Prompt Exploration with Large Language Models. *Proceedings of the 36th Annual ACM Symposium on User Interface Software and Technology*.
- [15] Zhan, J., Ai, Q., Liu, Y., Chen, J., & Ma, S. (2024). Capability-aware Prompt Reformulation Learning for Text-to-Image Generation. *Proceedings of the 47th International ACM SIGIR Conference on Research and Development in Information Retrieval*.
- [16] Xie, Y., Pan, Z., Ma, J., Luo, J., & Mei, Q. (2023). A Prompt Log Analysis of Text-to-Image Generation Systems. *Proceedings of the ACM Web Conference 2023*.
- [17] Godase, M. V., Mulani, A., Ghodak, M. R., Birajadar, M. G., Takale, M. S., & Kolte, M. A MapReduce and Kalman Filter based Secure IIoT Environment in Hadoop. *Sanshodhak*, Volume 19, June 2024.
- [18] Mulani, A. O., & Mane, P. B. (2017). Watermarking and cryptography based image authentication on reconfigurable platform. *Bulletin of Electrical Engineering and Informatics*, 6(2), 181-187.
- [19] Gadade, B., Mulani, A. O., & Harale, A. D. IoT Based Smart School Bus and Student Tracking System. *Sanshodhak*, Volume 19, June 2024.
- [20] Dhanawadel, A., Mulani, A. O., & Pise, A. C. IOT based Smart farming using Agri BOT. *Sanshodhak*, Volume 20, June 2024.
- [21] Mulani, A., & Mane, P. B. (2016). *DWT based robust invisible watermarking*. Scholars' Press.
- [22] R. G. Ghodke, G. B. Birajdar, A.O. Mulani, G.N. Shinde, R.B. Pawar, Design and Development of an Efficient and Cost-Effective surveillance Quadcopter using Arduino, *Sanshodhak*, Volume 20, June 2024.
- [23] R. G. Ghodke, G. B. Birajdar, A.O. Mulani, G.N. Shinde, R.B. Pawar, Design and Development of Wireless Controlled ROBOT using Bluetooth Technology, *Sanshodhak*, Volume 20, June 2024.
- [24] Swami, S. S., & Mulani, A. O. (2017, August). An efficient FPGA implementation of discrete wavelet transform for image compression. In *2017 International Conference on Energy, Communication, Data Analytics and Soft Computing (ICECDS)* (pp. 3385-3389). IEEE.
- [25] Mane, P. B., & Mulani, A. O. (2018). High speed area efficient FPGA implementation of AES algorithm. *International Journal of Reconfigurable and Embedded Systems*, 7(3), 157-165.
- [26] Mulani, A. O., & Mane, P. B. (2016). Area efficient high speed FPGA based invisible watermarking for image authentication. *Indian journal of Science and Technology*, 9(39), 1-6.
- [27] Kashid, M. M., Karande, K. J., & Mulani, A. O. (2022, November). IoT-based environmental parameter monitoring using machine learning approach. In *Proceedings of the International Conference on Cognitive and Intelligent Computing: ICCIC 2021, Volume 1* (pp. 43-51). Singapore: Springer Nature Singapore.



- [28] Nagane, U. P., & Mulani, A. O. (2021). Moving object detection and tracking using Matlab. *Journal of Science and Technology*, 6(1), 2456-5660.
- [29] Kulkarni, P. R., Mulani, A. O., & Mane, P. B. (2016). Robust invisible watermarking for image authentication. In *Emerging Trends in Electrical, Communications and Information Technologies: Proceedings of ICECIT-2015* (pp. 193-200). Singapore: Springer Singapore.
- [30] Ghodake, M. R. G., & Mulani, M. A. (2016). Sensor based automatic drip irrigation system. *Journal for Research*, 2(02).
- [31] Mandwale, A. J., & Mulani, A. O. (2015, January). Different Approaches For Implementation of Viterbi decoder on reconfigurable platform. In *2015 International Conference on Pervasive Computing (ICPC)* (pp. 1-4). IEEE.
- [32] Jadhav, M. M., Chavan, G. H., & Mulani, A. O. (2021). Machine learning based autonomous fire combat turret. *Turkish Journal of Computer and Mathematics Education*, 12(2), 2372-2381.
- [33] Shinde, G., & Mulani, A. (2019). A robust digital image watermarking using DWT-PCA. *International Journal of Innovations in Engineering Research and Technology*, 6(4), 1-7.
- [34] Mane, D. P., & Mulani, A. O. (2019). High throughput and area efficient FPGA implementation of AES algorithm. *International Journal of Engineering and Advanced Technology*, 8(4).
- [35] Mulani, A. O., & Mane, D. P. (2017). An Efficient implementation of DWT for image compression on reconfigurable platform. *International Journal of Control Theory and Applications*, 10(15), 1-7.
- [36] Deshpande, H. S., Karande, K. J., & Mulani, A. O. (2015, April). Area optimized implementation of AES algorithm on FPGA. In *2015 International Conference on Communications and Signal Processing (ICCSP)* (pp. 0010-0014). IEEE.
- [37] Deshpande, H. S., Karande, K. J., & Mulani, A. O. (2014, April). Efficient implementation of AES algorithm on FPGA. In *2014 International Conference on Communication and Signal Processing* (pp. 1895-1899). IEEE.
- [38] Kulkarni, P., & Mulani, A. O. (2015). Robust invisible digital image watermarking using discrete wavelet transform. *International Journal of Engineering Research & Technology (IJERT)*, 4(01), 139-141.
- [39] Mulani, A. O., Jadhav, M. M., & Seth, M. (2022). Painless Non-invasive blood glucose concentration level estimation using PCA and machine learning. *The CRC Book entitled Artificial Intelligence, Internet of Things (IoT) and Smart Materials for Energy Applications*.
- [40] Mulani, A. O., & Shinde, G. N. (2021). An approach for robust digital image watermarking using DWT-PCA. *Journal of Science and Technology*, 6(1).
- [41] Mulani, A. O., & Mane, P. B. (2014, October). Area optimization of cryptographic algorithm on less dense reconfigurable platform. In *2014 International Conference on Smart Structures and Systems (ICSSS)* (pp. 86-89). IEEE.
- [42] Jadhav, H. M., Mulani, A., & Jadhav, M. M. (2022). Design and development of chatbot based on reinforcement learning. *Machine Learning Algorithms for Signal and Image Processing*, 219-229.
- [43] Mulani, A. O., & Mane, P. (2018). Secure and area efficient implementation of digital image watermarking on reconfigurable platform. *International Journal of Innovative Technology and Exploring Engineering*, 8(2), 56-61.
- [44] Kalyankar, P. A., Mulani, A. O., Thigale, S. P., Chavhan, P. G., & Jadhav, M. M. (2022). Scalable face image retrieval using AESC technique. *Journal Of Algebraic Statistics*, 13(3), 173-176.
- [45] Takale, S., & Mulani, A. (2022). DWT-PCA based video watermarking. *Journal of Electronics, Computer Networking and Applied Mathematics (JECNAM)* ISSN, 2799-1156.
- [46] Kamble, A., & Mulani, A. O. (2022). Google assistant based device control. *Int. J. of Aquatic Science*, 13(1), 550-555.
- [47] Kondekar, R. P., & Mulani, A. O. (2017). Raspberry Pi based voice operated Robot. *International Journal of Recent Engineering Research and Development*, 2(12), 69-76.



- [48] Ghodake, R. G., & Mulani, A. O. (2018). Microcontroller based automatic drip irrigation system. In *Techno-Societal 2016: Proceedings of the International Conference on Advanced Technologies for Societal Applications* (pp. 109-115). Springer International Publishing.
- [49] Mulani, A. O., Birajadar, G., Ivković, N., Salah, B., & Darlis, A. R. (2023). Deep learning based detection of dermatological diseases using convolutional neural networks and decision trees. *Traitement du Signal*, 40(6), 2819.
- [50] Boxey, A., Jadhav, A., Gade, P., Ghanti, P., & Mulani, A. O. (2022). Face Recognition using Raspberry Pi. *Journal of Image Processing and Intelligent Remote Sensing (JIPIRS) ISSN*, 2815-0953.
- [51] Patale, J. P., Jagadale, A. B., Mulani, A. O., & Pise, A. (2023). A Systematic survey on Estimation of Electrical Vehicle. *Journal of Electronics, Computer Networking and Applied Mathematics (JECNAM) ISSN*, 2799-1156.
- [52] Gadade, B., & Mulani, A. (2022). Automatic System for Car Health Monitoring. *International Journal of Innovations in Engineering Research and Technology*, 57-62.
- [53] Shinde, M. R. S., & Mulani, A. O. (2015). Analysis of Biomedical Image Using Wavelet Transform. *International Journal of Innovations in Engineering Research and Technology*, 2(7), 1-7.
- [54] Mandwale, A., & Mulani, A. O. (2014, December). Implementation of convolutional encoder & different approaches for viterbi decoder. In *IEEE International Conference on Communications, Signal Processing Computing and Information technologies*.
- [55] Mulani, A. O., Jadhav, M. M., & Seth, M. (2022). Painless machine learning approach to estimate blood glucose level with non-invasive devices. In *Artificial intelligence, internet of things (IoT) and smart materials for energy applications* (pp. 83-100). CRC Press.
- [56] Maske, Y., Jagadale, A. B., Mulani, A. O., & Pise, A. C. (2023). Development of BIOBOT system to assist COVID patient and caretakers. *European Journal of Molecular & Clinical Medicine*, 10(01), 2023.
- [57] Utpat, V. B., Karande, D. K., & Mulani, D. A. Grading of Pomegranate Using Quality Analysis. *International Journal for Research in Applied Science & Engineering Technology (IJRASET)*, 10.
- [58] Takale, S., & Mulani, D. A. (2022). Video Watermarking System. *International Journal for Research in Applied Science & Engineering Technology (IJRASET)*, 10.
- [59] Mandwale, A., & Mulani, A. O. (2015, January). Different approaches for implementation of Viterbi decoder. In *IEEE international conference on pervasive computing (ICPC)*.
- [60] Maske, Y., Jagadale, M. A., Mulani, A. O., & Pise, A. (2021). Implementation of BIOBOT System for COVID Patient and Caretakers Assistant Using IOT. *International Journal of Information Technology and*, 30-43.
- [61] Mulani, A. O., & Mane, D. P. (2016). Fast and Efficient VLSI Implementation of DWT for Image Compression. *International Journal for Research in Applied Science & Engineering Technology*, 5, 1397-1402.
- [62] Kambale, A. (2023). Home automation using google assistant. *UGC care approved journal*, 32(1), 1071-1077.
- [63] Pathan, A. N., Shejal, S. A., Salgar, S. A., Harale, A. D., & Mulani, A. O. (2022). Hand gesture controlled robotic system. *Int. J. of Aquatic Science*, 13(1), 487-493.
- [64] Korake, D. M., & Mulani, A. O. (2016). Design of Computer/Laptop Independent Data transfer system from one USB flash drive to another using ARM11 processor. *International Journal of Science, Engineering and Technology Research*.
- [65] Mandwale, A., & Mulani, A. O. (2016). Implementation of High Speed Viterbi Decoder using FPGA. *International Journal of Engineering Research & Technology, IJERT*.
- [66] Kolekar, S. D., Walekar, V. B., Patil, P. S., Mulani, A. O., & Harale, A. D. (2022). Password Based Door Lock System. *Int. J. of Aquatic Science*, 13(1), 494-501.
- [67] Shinde, R., & Mulani, A. O. (2015). Analysis of Biomedical Image. *International Journal on Recent & Innovative trend in technology (IJRITT)*.



- [68] Sawant, R. A., & Mulani, A. O. (2022). Automatic PCB Track Design Machine. *International Journal of Innovative Science and Research Technology*, 7(9).
- [69] ABHANGRAO, M. R., JADHAV, M. S., GHODKE, M. P., & MULANI, A. (2017). Design And Implementation Of 8-bit Vedic Multiplier. *International Journal of Research Publications in Engineering and Technology (ISSN No: 2454-7875)*.
- [70] Gadade, B., Mulani, A. O., & Harale, A. D. (2024). Iot based smart school bus and student monitoring system. *Naturalista Campano*, 28(1), 730-737.
- [71] Mulani, D. A. O. (2024). A Comprehensive Survey on Semi-Automatic Solar-Powered Pesticide Sprayers for Farming. *Journal of Energy Engineering and Thermodynamics (JEET) ISSN*, 2815-0945.
- [72] Salunkhe, D. S. S., & Mulani, D. A. O. (2024). Solar Mount Design Using High-Density Polyethylene. *NATURALISTA CAMPANO*, 28(1).
- [73] Seth, M. (2022). Painless Machine learning approach to estimate blood glucose level of Non-Invasive device. *Artificial Intelligence, Internet of Things (IoT) and Smart Materials for Energy Applications*.
- [74] Kolhe, V. A., Pawar, S. Y., Gohery, S., Mulani, A. O., Sundari, M. S., Kiradoo, G., ... & Sunil, J. (2024). Computational and experimental analyses of pressure drop in curved tube structural sections of Coriolis mass flow metre for laminar flow region. *Ships and Offshore Structures*, 19(11), 1974-1983.
- [75] Basawaraj Birajadar, G., Osman Mulani, A., Ibrahim Khalaf, O., Farhah, N., G Gawande, P., Kinage, K., & Abdullah Hamad, A. (2024). Epilepsy identification using hybrid CoPrO-DCNN classifier. *International Journal of Computing and Digital Systems*, 16(1), 783-796.
- [76] Kedar, M. S., & Mulani, A. (2021). IoT Based Soil, Water and Air Quality Monitoring System for Pomegranate Farming. *Journal of Electronics, Computer Networking and Applied Mathematics (JECNAM) ISSN*, 2799-1156.
- [77] Godse, A. P. A.O. Mulani (2009). *Embedded Systems (First Edition)*.
- [78] Pol, R. S., Bhalerao, M. V., & Mulani, A. O. A real time IoT based System Prediction and Monitoring of Landslides. *International Journal of Food and Nutritional Sciences*, Volume 11, Issue 7, 2022.
- [79] Mulani, A. O., Sardey, M. P., Kinage, K., Salunkhe, S. S., Fegade, T., & Fegade, P. G. (2025). ML-powered Internet of Medical Things (MLIOMT) structure for heart disease prediction. *Journal of Pharmacology and Pharmacotherapeutics*, 16(1), 38-45.
- [80] Aiwale, S., Kolte, M. T., Harpale, V., Bendre, V., Khurge, D., Bhandari, S., ... & Mulani, A. O. (2024). Non-invasive Anemia Detection and Prediagnosis. *Journal of Pharmacology and Pharmacotherapeutics*, 15(4), 408-416.
- [81] Mulani, A. O., Bang, A. V., Birajadar, G. B., Deshmukh, A. B., Jadhav, H. M., & Liyakat, K. K. S. (2024). IoT Based Air, Water, and Soil Monitoring System for Pomegranate Farming. *Annals of Agri-Bio Research*, 29(2), 71-86.
- [82] Kulkarni, T. M., & Mulani, A. O. (2024). Face Mask Detection on Real Time Images and Videos using Deep Learning. *International Journal of Electrical Machine Analysis and Design (IJEMAD)*, 2(1).
- [83] Thigale, S. P., Jadhav, H. M., Mulani, A. O., Birajadar, G. B., Nagrale, M., & Sardey, M. P. (2024). Internet of things and robotics in transforming healthcare services. *Afr J Biol Sci (S Afr)*, 6(6), 1567-1575.
- [84] Pol, D. R. S. (2021). Cloud Based Memory Efficient Biometric Attendance System Using Face Recognition. *Stochastic Modeling & Applications*, 25(2).
- [85] Nagtilak, M. A. G., Ulegaddi, M. S. N., Adat, M. A. S., & Mulani, A. O. (2021). Breast Cancer Prediction using Machine Learning.
- [86] Rahul, G. G., & Mulani, A. O. (2016). Microcontroller Based Drip Irrigation System.
- [87] Kulkarni, T. M., & Mulani, A. O. Deep Learning Based Face-Mask Detection: An Approach to Reduce Pandemic Spreads in Human Healthcare. *African Journal of Biological Sciences*, 6(6), 2024.
- [88] Mulani, A., & Mane, P. B. (2016). *DWT based robust invisible watermarking*. Scholars' Press.



- [89] Dr. Vaishali Satish Jadhav, Dr. Shweta Sadanand Salunkhe, Dr. Geeta Salunkhe, Pranali Rajesh Yawle, Dr. Rahul S. Pol, Dr. Altaf Osman Mulani, Dr. Manish Rana, Iot Based Health Monitoring System for Human, Afr. J. Biomed. Res. Vol. 27 (September 2024).
- [90] Dr. Vaishali Satish Jadhav, Geeta D. Salunke, Kalyani Ramesh Chaudhari, Dr. Altaf Osman Mulani, Dr. Sampada Padmakar Thigale, Dr. Rahul S. Pol, Dr. Manish Rana, Deep Learning-Based Face Mask Recognition in Real-Time Photos and Videos, Afr. J. Biomed. Res. Vol. 27 (September 2024).
- [91] Altaf Osman Mulani, Electric Vehicle Parameters Estimation Using Web Portal, Recent Trends in Electronics & Communication Systems, Volume 10, Issue 3, 2023.
- [92] Aryan Ganesh Nagtilak, Sneha Nitin Ulegaddi, Mahesh Mane, Altaf O. Mulani, Automatic Solar Powered Pesticide Sprayer for Farming, International Journal of Microwave Engineering and Technology, Volume 9 No. 2, 2023.
- [93] Annasaheb S. Dandage, Vitthal R. Rupnar, Tejas A Pise, and A. O. Mulani, Real-Time Language Translation Application Using Tkinter. International Journal of Digital Communication and Analog Signals. 2025; 11(01): -p.
- [94] AnnaSaheb S Dandage, Vitthal R. Rupnar, Tejas A Pise, and A. O. Mulani, IoT-Powered Weather Monitoring and Irrigation Automation: Transforming Modern Farming Practices. . 2025; 11(01): -p.
- [95] Mulani, A.O., Kulkarni, T.M. (2025). Face Mask Detection System Using Deep Learning: A Comprehensive Survey. In: Singh, S., Arya, K.V., Rodriguez, C.R., Mulani, A.O. (eds) Emerging Trends in Artificial Intelligence, Data Science and Signal Processing. AIDSP 2023. Communications in Computer and Information Science, vol 2439. Springer, Cham. https://doi.org/10.1007/978-3-031-88759-8_3.
- [96] Karve, S., Gangonda, S., Birajadar, G., Godase, V., Ghodake, R., Mulani, A.O. (2025). Optimized Neural Network for Prediction of Neurological Disorders. In: Singh, S., Arya, K.V., Rodriguez, C.R., Mulani, A.O. (eds) Emerging Trends in Artificial Intelligence, Data Science and Signal Processing. AIDSP 2023. Communications in Computer and Information Science, vol 2440. Springer, Cham. https://doi.org/10.1007/978-3-031-88762-8_18.
- [97] Saurabh Singh, Karm Veer Arya, Ciro Rodriguez Rodriguez, and Altaf Osman Mulani, Emerging Trends in Artificial Intelligence, Data Science and Signal Processing, Communications in Computer and Information Science (CCIS), volume 2440.
- [98] Saurabh Singh, Karm Veer Arya, Ciro Rodriguez Rodriguez, and Altaf Osman Mulani, Emerging Trends in Artificial Intelligence, Data Science and Signal Processing, Communications in Computer and Information Science (CCIS), volume 2439.
- [99] Godase, V., Mulani, A., Pawar, A., & Sahani, K. (2025). A Comprehensive Review on PIR Sensor-Based Light Automation Systems. International Journal of Image Processing and Smart Sensors, 1(1), 22-29.
- [100] Godase, V., Mulani, A., Takale, S., & Ghodake, R. (2025). Comprehensive Review on Automated Field Irrigation using Soil Image Analysis and IoT. Journal of Advance Electrical Engineering and Devices, 3(1), 46-55.
- [101] Altaf Osman Mulani, Deshmukh M., Jadhav V., Chaudhari K., Mathew A.A., Shweta Salunkhe. Transforming Drug Therapy with Deep Learning: The Future of Personalized Medicine. Drug Research. 2025 Aug 29.
- [102] Altaf O. Mulani, Vaibhav V. Godase, Swapnil R. Takale, Rahul G. Ghodake (2025), Image Authentication Using Cryptography and Watermarking, International Journal of Image Processing and Smart Sensors, Vol. 1, Issue 2, pp 27-34.
- [103] Altaf O. Mulani, Vaibhav V. Godase, Swapnil R. Takale, Rahul G. Ghodake (2025), Advancements in Artificial Intelligence: Transforming Industries and Society, International Journal of Artificial Intelligence of Things (AIoT) in Communication Industry, Vol. 1, Issue 2, pp 1-5.
- [104] Altaf O. Mulani, Vaibhav V. Godase, Swapnil R. Takale, Rahul G. Ghodake (2025), AI-Powered Predictive Analytics in Healthcare: Revolutionizing Disease Diagnosis and Treatment, Journal of Advance Electrical Engineering and Devices, Vol. 3, Issue 2, pp 27-34.



- [105] Godase, V., Mulani, A., Takale, S., & Ghodake, R. (2025). A Holistic Review of Automatic Drip Irrigation Systems: Foundations and Emerging Trends. *Available at SSRN 5247778*.
- [106] V. Godase, R. Ghodake, S. Takale, and A. Mulani, —Design and Optimization of Reconfigurable Microwave Filters Using AI Techniques, *International Journal of RF and Microwave Communication Technologies*, vol. 2, no. 2, pp.26–41, Aug. 2025.
- [107] V. Godase, A. Mulani, R. Ghodake, S. Takale, “Automated Water Distribution Management and Leakage Mitigation Using PLC Systems,” *Journal of Control and Instrumentation Engineering*, vol.11, no. 3, pp. 1-8, Aug. 2025.
- [108] V. Godase, A. Mulani, R. Ghodake, S. Takale, “PLC-Assisted Smart Water Distribution with Rapid Leakage Detection and Isolation,” *Journal of Control Systems and Converters*, vol. 1, no. 3, pp. 1-13, Aug. 2025.
- [109] V. V. Godase, S. R. Takale, R. G. Ghodake, and A. Mulani, “Attention Mechanisms in Semantic Segmentation of Remote Sensing Images,” *Journal of Advancement in Electronics Signal Processing*, vol. 2, no. 2, pp. 45–58, Aug. 2025.
- [110] D. Waghmare, A. Mulani, S. R. Takale, V. Godase, and A. Mulani, “A Comprehensive Review on Automatic Fruit Sorting and Grading Techniques with Emphasis on Weight-based Classification,” *Research & Review: Electronics and Communication Engineering*, vol. 2, no. 3, pp. 1-10, Oct. 2025.
- [111] Karande, K. J., & Talbar, S. N. (2014). Independent component analysis of edge information for face recognition. Springer India.
- [112] Karande, K. J., & Talbar, S. N. (2008). Face recognition under variation of pose and illumination using independent component analysis. ICGST-GVIP, ISSN.
- [113] Gaikwad, D. S., & Karande, K. J. (2016). Image processing approach for grading and identification of diseases on pomegranate fruit: An overview. *International Journal of Computer Science and Information Technologies*, 7, 519-522.
- [114] Kawathekar, P. P., & Karande, K. J. (2014, July). Severity analysis of Osteoarthritis of knee joint from X-ray images: A Literature review. In 2014 International Conference on Signal propagation and computer technology (ICSPCT 2014) (pp. 648-652). IEEE.
- [115] Daithankar, M. V., Karande, K. J., & Harale, A. D. (2014, April). Analysis of skin color models for face detection. In 2014 International Conference on Communication and Signal Processing (pp. 533-537). IEEE.
- [116] Karande, J. K., Talbar, N. S., & Inamdar, S. S. (2012, May). Face recognition using oriented Laplacian of Gaussian (OLOG) and independent component analysis (ICA). In 2012 Second International Conference on Digital Information and Communication Technology and it's Applications (DICTAP) (pp. 99-103). IEEE.
- [117] Shubham Salunkhe, Pruthviraj Zambare, Sakshi Shinde, S. K. Godase. (2024). API Development for Cloud Parameter Curation International. *Journal of Electrical and Communication Engineering Technology*, 2(1). <https://doi.org/10.37591/ijecet>
- [118] Badave, A., Pawale, A., Andhale, T., Godase, S. K., & STM JOURNALS. (2024). Smart home safety using fire and gas detection system. *Recent Trends in Fluid Mechanics*, 1, 35–43. <https://journals.stmjournals.com/rtfm>
- [119] Asabe, H., Asabe, R., Lengare, O., & Godase, S. (2025). IOT- BASED STORAGE SYSTEM FOR MANAGING VOLATILE MEDICAL RESOURCES IN HEALTHCARE FACILITIES. *INTERNATIONAL JOURNAL OF PROGRESSIVE RESEARCH IN ENGINEERING MANAGEMENT AND SCIENCE (IJPREMS)*, 05(03), 2427–2433. <https://www.ijprems.com>
- [120] Karche, S. N., Mulani, A. O., Department of Electronics, SKN Sinhgad College of Engineering, Korti, & University of Solapur, Maharashtra, India. (2018). AESC Technique for Scalable Face Image Retrieval. *International Journal of Innovative Research in Computer and Communication Engineering*, 6(4), 3404–3405. <https://doi.org/10.15680/IJRCCE.2018.0604036>
- [121] Bankar, A. S., Harale, A. D., & Karande, K. J. (2021). Gestures Controlled Home Automation using Deep Learning: A Review. *International Journal of Current Engineering and Technology*, 11(06), 617–621.



<https://doi.org/10.14741/ijcet/v.11.6.4>

- [122] Mali, A. S., Ghadge, S. K., Adat, A. S., & Karande, S. V. (2024). Intelligent Medication Management System. *IJSRD - International Journal for Scientific Research & Development*, Vol. 12(Issue 3).
- [123] Water Level Control, Monitoring and Altering System by using GSM in Irrigation Based on Season. (2019). In *International Research Journal of Engineering and Technology (IRJET)* (Vol. 06, Issue 04, p. 1035) [Journal-article]. <https://www.irjet.net>
- [124] Modi, S., Misal, V., Kulkarni, S., & Mali A.S. (2025). Hydroponic Farming Monitoring System - Automated system to monitor and control nutrient and pH levels. In *Journal of Microcontroller Engineering and Applications* (Vol. 12, Issue 3, pp. 11–16). <https://doi.org/10.37591/JoMEA>
- [125] Siddheshwar S. Gangonda, Prashant P. Patavardhan, Kailash J. Karande, “VGHN: variations aware geometric moments and histogram features normalization for robust uncontrolled face recognition”, *International Journal of Information Technology*, <https://doi.org/10.1007/s41870-021-00703-0>.
- [126] Siddheshwar Gangonda and Prachi Mukherji, “Speech Processing for Marathi Numeral Recognition using MFCC & DTW Features”, *International Journal of Engineering Research And Applications (IJERA)* pp. 118-122, ISSN: 2248-9622.
- [127] Siddheshwar S. Gangonda, Prashant P. Patavardhan, Kailash J. Karande, “Recognition of Marathi Numerals Using MFCC and DTW Features”, *Book Title: Recent Trends on Image Processing and Pattern Recognition, RTIP2R 2018, CCIS 1037*, pp. 1–11, © Springer Nature Singapore Pte Ltd. 2019 https://doi.org/10.1007/978-981-13-9187-3_17.
- [128] Siddheshwar S. Gangonda, Prashant P. Patavardhan, Kailash J. Karande, “Analysis of Face Recognition Algorithms for Uncontrolled Environments”, *Book Title: Computing, Communication and Signal Processing*, pp. 919–926, © Springer Nature Singapore Pte Ltd. 2018.
- [129] Siddheshwar S. Gangonda, Prashant P. Patavardhan, Kailash J. Karande, “Recognition of Marathi Numerals using MFCC and DTW Features”, 2nd International Conference on Recent Trends in Image Processing and Pattern Recognition (RTIP2R 2018), 21th -22th Dec., 2018, organized by Solapur University, Solapur in collaboration with University of South Dakota (USA) and Universidade de Evora (Portugal), India.
- [130] Siddheshwar S. Gangonda, Prashant P. Patavardhan, Kailash J. Karande, “A Comprehensive Survey of Face Databases for Constrained and Unconstrained Environments”, 2nd IEEE Global Conference on Wireless Computing & Networking (GCWCN-2018), 23th-24th Nov., 2018, organized by STES's Sinhgad Institute of Technology, Lonavala, India.
- [131] Siddheshwar S. Gangonda, Prashant P. Patavardhan, Kailash J. Karande, “An Extensive Survey of Prominent Researches in Face Recognition under different Conditions”, 4th International Conference on Computing, Communication, Control And Automation (ICCUBEA-2018), 16th to 18th Aug. 2018 organized by Pimpri Chinchwad College of Engineering (PCCOE), Pune, India.
- [132] Siddheshwar S. Gangonda, Prashant P. Patavardhan, Kailash J. Karande, “Analysis of Face Recognition Algorithms for Uncontrolled Environments”, 3rd International Conference on Computing, Communication and Signal Processing (ICASP 2018), 26th-27th Jan.2018, organized by Dr. BATU, Lonere, India.
- [133] Siddheshwar Gangonda and Prachi Mukherji, “Speech Processing for Marathi Numeral Recognition”, International Conference on Recent Trends, Feb 2012, IOK COE, Pune.
- [134] S. S. Gangonda, “Bidirectional Visitor Counter with automatic Door Lock System”, National Conference on Computer, Communication and Information Technology (NCCCIT-2018), 30th and 31st March 2018 organized by Department of Electronics and Telecommunication Engineering, SKN SCOE, Korti, Pandharpur.
- [135] Siddheshwar Gangonda and Prachi Mukherji, “Speech Processing for Marathi Numeral Recognition using MFCC & DTW Features”, ePGCON 2012, 23rd and 24th April 2012 organized by Commis COE for Woman, Pune.



- [136] Siddheshwar Gangonda and Prachi Mukherji, "Speech Processing for Marathi Numeral Recognition", National Conference on Emerging Trends in Engineering and Technology (VNCET'12), 30th March 2012 organized by Vidyavardhini's College of Engineering and Technology, Vasai Road, Thane.
- [137] Siddheshwar Gangonda and Prachi Mukherji, "Speech Processing for Marathi Numeral Recognition", ePGCON 2011, 26th April 2011 organized by MAEER's MIT, Kothrud, Pune-38.
- [138] Siddheshwar Gangonda, "Medical Image Processing", Aavishkar-2K7, 17th and 18th March 2007 organized by Department of Electronics and Telecommunication Engineering, SVERI's COE, Pandharpur.
- [139] Siddheshwar Gangonda, "Image enhancement & Denoising", VISION 2k7, 28th Feb-2nd March 2007 organized by M.T.E. Society's Walchand College of Engineering, Sangli.
- [140] Siddheshwar Gangonda, "Electromagnetic interference & compatibility" KSHITIJ 2k6, 23rd and 24th Sept. 2006 organized by Department of Mechanical Engineering, SVERI's COE, Pandharpur.
- [141] A. Pise and K. Karande, "A genetic Algorithm-Driven Energy-Efficient routing strategy for optimizing performance in VANETs," Engineering Technology and Applied Science Research, vol. 15, no. 5, 2025, [Online]. Available: <https://etasr.com/index.php/ETASR/article/view/12744>
- [142] A. C. Pise, K. J. Karande, "Investigating Energy-Efficient Optimal Routing Protocols for VANETs: A Comprehensive Study", ICT for Intelligent Systems, Lecture Notes in Networks and Systems 1109, Proceedings of ICTIS 2024 Volume 3, Lecture Notes in Networks and Systems, Springer, Singapore, ISSN 2367-3370, PP 407-417, 29 October 2024 https://doi.org/10.1007/978-981-97-6675-8_33.
- [143] A. C. Pise, et. al., "Smart Vehicle: A Systematic Review", International Journal - The Ciência & Engenharia - Science & Engineering Journal ISSN: 0103-944X Volume 11 Issue 1, 2023pp: 992-998, 2023.
- [144] A. C. Pise, et. al., "Smart Vehicle: A Systematic Review", International Journal of Research Publication and Reviews, ISSN 2582-7421, Vol 4, no 10, pp 2728-2731 October 2023.
- [145] A. C. Pise, et. al., "Development of BIOBOT System to Assist COVID Patient and Caretakers", European Journal of Molecular and Clinical Medicine ; 10(1):3472-3480, 2023.
- [146] A. C. Pise, et. al., "IoT Based Landmine Detection Robot", International Journal of Research in Science & Engineering ISSN: 2394-8299 Vol: 03, No. 04, June-July 2023.
- [147] A. C. Pise, et. al., "A Systematic survey on Estimation of Electrical Vehicle", Journal of Electronics, Computer Networking and Applied Mathematics (JECNAM) ISSN: 2799-1156, Volume 3, Issue 01, Pages 1-6, December 2023.
- [148] A. C. Pise, et. al., "Python Algorithm to Estimate Range of Electrical Vehicle", Web of Science, Vol 21, No 1 (2022) December 2022
- [149] A. C. Pise, et. al., "Implementation of BIOBOT System for COVID Patient and Caretakers Assistant using IOT", International Journal of Information technology and Computer Engineering. 30-43. 10.55529/ijitc.21.30.43, (2022).
- [150] A. C. Pise, et. al., "An IoT Based Real Time Monitoring of Agricultural and Micro irrigation system", International journal of scientific research in Engineering and management (IJSREM), VOLUME: 06 ISSUE: 04 | APRIL – 2022, ISSN:2582-3930.
- [151] A. C. Pise, Dr. K. J. Karande, "An Exploratory study of Cluster Based Routing Protocol in VANET: A Review", International Journal of Advanced Research in Engineering and Technology(IJARET), 12,10, 2021, 17-30, Manuscript ID :00000-94375 Source ID : 00000006, Journal_uploads/IJARET/VOLUME_12_ISSUE_10/IJARET_12_10_002.pdf
- [152] A. C. Pise, et. al., "Android based Portable Health Support System," A Peer Referred & Indexed International Journal of Research, Vol.8, issue.4 , April 2019.
- [153] A. C. Pise, et. al., "Facial Expression Recognition Using Image Processing," International Journal of VLSI Design, Microelectronics and Embedded System, Vol.3, issue.2 , July 2018.
- [154] A. C. Pise, et. al., "Detection of Cast Iron Composition by Cooling Curve Analysis using Thermocouple Temperature Sensor," UGC Approved International Journal of Academic Science (IJRECE), Vol.6, Issue.3 , July-September 2018.



- [155] A. C. Pise, et. al., "Android Based Portable Health Support", System International Journal of Engineering Sciences & Research Technology (IJESRT 2017) Vol.6, Issue 8, pp 85-88 5th Aug 2017
- [156] A. C. Pise, et. al., "Adaptive Noise Cancellation in Speech Signal", International Journal of Innovative Engg and Technology, 2017
- [157] A. C. Pise, et. al., "Lung Cancer Detection System by using Baysian Classifier", ISSN 2454-7875, IJRPET, published online in conference special issue VESCOMM-2016, February 2016
- [158] A. C. Pise, et. al., "Review on Agricultural Plant Diseases Detection by Image Processing", ISSN 2278-621X, IJLTET, Vol 7, Issue 1 May 2016
- [159] A. C. Pise, et. al. "Segmentation of Retinal Images for Glaucoma Detection", International Journal of Engineering Research and Technology (06, June-2015).
- [160] A. C. Pise, et. al. "Color Local Texture Features Based Face Recognition", International Journal of Innovations in Engineering and Technology(IJIET) , Dec. 2014
- [161] A. C. Pise, et. al. "Single Chip Solution For Multimode Robotic Control", International Journal of Engineering Research and Technology (IJERT-2014), Vol. 3, Issue 12, Dec. 2014.
- [162] Anjali C. Pise et. al., "Remote monitoring of Greenhouse parameters using zigbee Wireless Sensor Network", International Journal of Engineering Research & Technology ISSN 2278-0181 (online) Vol. 3, Issue 2, and pp: (2412-2414), Feb. 2014.
- [163] A. C. Pise, K. J. Karande, "Cluster Head Selection Based on ACO In Vehicular Ad-hoc Networks", Machine Learning for Environmental Monitoring in Wireless Sensor Networks
- [164] A. C. Pise, K. J. Karande, "Architecture, Characteristics, Applications and Challenges in Vehicular Ad Hoc Networks" Presented in 27th IEEE International Symposium on Wireless Personal Multimedia Communications (WPMC 2024) "Secure 6G – AI Nexus: Where Technology Meets Humanity" Accepted for book chapter to be published in international Scopus index book by River publisher.
- [165] A. C. Pise, Dr. K. J. Karande, "K-mean Energy Efficient Optimal Cluster Based Routing Protocol in Vehicular Ad Hoc Networks", International Conference on Innovations in Artificial Intelligence and Machine Learning (ICAIML-2022), August 20th and 21st 2022 Springer database Conference.
- [166] A. C. Pise, Mr. D. Nale, "Web-Based Application for Result Analysis", ", International Conference on Innovations in Artificial Intelligence and Machine Learning (ICAIML-2022), August 20th and 21st 2022 Springer database Conference.
- [167] A. C. Pise, et. al., "Detection of Cast Iron Composition by Cooling Curve Analysis using Thermocouple Temperature Sensor," 2nd International Conference on Engineering Technology, Science and Management Innovation (ICETSMI – 2018), 2nd September 2018.
- [168] A. C. Pise, et. al., "Facial Expression Recognition Using Facial Features," IEEE International Conference on Communication and Electronics Systems (ICCES 2018), October 2018.
- [169] A. C. Pise, et. al., "Estimating Parameters of Cast Iron Composition using Cooling Curve Analysis," IEEE International Conference on Communication and Electronics Systems (ICCES 2018), Coimbatore, October 2018.
- [170] A. C. Pise, et. al., "Android based portable Health Support System," International Conference on Innovations in Engineering and Technology (CIET 2016), SKN Sinhgad College of Engineering, 30-31 Dec 2016.
- [171] A. C. Pise, et. al., "Baysian Classifier & FCM Segmentation for Lung Cancer Detection in early stage," International Conference on Innovations in Engineering and Technology (CIET 2016), SKN Sinhgad College of Engineering, 30-31 Dec 2016.
- [172] A. C. Pise, et. al., "Cast Iron Composition Measurement by Coding Curve Analysis," International Conference on Innovations in Engineering and Technology (CIET 2016), SKN Sinhgad College of Engineering, 30-31 Dec 2016.
- [173] A. C. Pise, et. al., "War field Intelligence Defence Flaging Vehicle," International Conference on Innovations in Engineering and Technology (CIET 2016), SKN Sinhgad College of Engineering, 30-31 Dec 2016.



- [174] A. C. Pise, et. al. "Disease Detection of Pomegranate Plant", IEEE sponsored International Conference on Computation of Power, Energy, Information and Communication, 22- 23 Apr. 2015.
- [175] A. C. Pise, P. Bankar. "Face Recognition by using GABOR and LBP", IEEE International Conference on Communication and Signal Processing, ICCSP, 2-4 Apr. 2015
- [176] A. C. Pise, et. al. "Single Chip Solution For Multimode Robotic Control", Ist IEEE International Conferece on Computing Communication and Automation, 26-27 Feb2015.
- [177] Anjali C. Pise, Vaishali S. Katti, "Efficient Design for Monitoring of Greenhouse Parameters using Zigbee Wireless Sensor Network", fifth SARC international conference IRF,IEEE forum ISBN 978-93-84209-21-6,pp 24-26, 25th May 2014
- [178] A. C. Pise, P. Bankar, "Face Recognition using Color Local Texture Features", International Conference on Electronics and Telecommunication, Electrical and Computer Engineering, Apr.2014.
- [179] A. C. Pise, et.al. "Monitoring parameters of Greenhouse using Zigbee Wireless Sensor Network", 1st International Conference on Electronics and Telecommunication, Electrical and Computer Engineering, 5-6 Apr.2014.
- [180] A. C. Pise, et. al. "Compensation schemes and performance Analysis of IQ Imbalances in Direct Conversion Receivers", International Conference at GHPCOE, Gujarat, (Online Proceeding is Available), 2009.
- [181] A. C. Pise, K. J. Karande, "Energy-Efficient Optimal Routing Protocols in VANETs", 66th Annual IETE Convention, AIC -2023 September16-17, 2023, under the Theme: The Role of 5G In Enabling Digital Transformation for Rural Upliftment.
- [182] A. C. Pise, et. al. "Automatic Bottle Filling Machine using Raspberry Pi", National Conference on computer ;Communication & information Technology (NCCIT-2018) dated 30th & 31st March 2018.
- [183] A. C. Pise, et. al. "Design & Implementation of ALU using VHDL", National Conference on computer ;Communication & information Technology (NCCIT-2018) dated 30th & 31st March 2018.
- [184] A. C. Pise, et. al. "Mechanism and Control of Autonomus four rotor Quad copter", National Conference on Computer, Electrical and Electronics Engineering, 23- 24 Apr. 2016.
- [185] A. C. Pise, et. al. "Segmentation of Optic Disk and Optic Cup from retinal Images", ICEECMPE – Chennai, June 2015
- [186] A. C. Pise, et. al. "Diseases Detection of Pomegranate Plant", IEEE Sponsored International conference on Computation of Power, Energy , April 2015.
- [187] A. C. Pise, et. al. "Compensation Techniques for I/Q Imbalance in Direct-Conversion Receivers", Conference at SCOE, Pune 2010.
- [188] A. C. Pise, et. al. "I/Q Imbalance compensation Techniques in Direct Conversion Receiver ", Advancing Trends in Engineering and Management Technologies, ATEMT-2009, Conference at Shri Ramdeobaba Kamla Nehru Engineering College, Nagpur, 20-21 November 2009
- [189] A. C. Pise, et. al. "Compensation Techniques for I/Q Imbalance in Direct Conversion Receiver", Conference at PICT, Pune 2008.
- [190] A. C. Pise, et. al. "I/Q Imbalance compensation Techniques in Direct Conversion Receiver ", Conference at DYCOE, Pune 2008.
- [191] A. C. Pise, et. al. "DUCHA: A New Dual channel MAC protocol for Multihop Ad-Hoc Networks", Conference at SVCP, Pune 2007.
- [192] Godase, V., Pawar, P., Nagane, S., & Kumbhar, S. (2024). Automatic railway horn system using node MCU. Journal of Control & Instrumentation, 15(1).
- [193] Godase, V., & Godase, J. (2024). Diet prediction and feature importance of gut microbiome using machine learning. Evolution in Electrical and Electronic Engineering, 5(2), 214-219.
- [194] Jamadade, V. K., Ghodke, M. G., Katakdhond, S. S., & Godase, V. A Comprehensive Review on Scalable Arduino Radar Platform for Real-time Object Detection and Mapping.
- [195] Godase, V. (2025). A comprehensive study of revolutionizing EV charging with solar-powered wireless solutions. Advance Research in Power Electronics and Devices e-ISSN, 3048-7145.



- [196] Godase, V. (2025, April). Advanced Neural Network Models for Optimal Energy Management in Microgrids with Integrated Electric Vehicles. In Proceedings of the International Conference on Trends in Material Science and Inventive Materials (ICTMIM-2025) DVD Part Number: CFP250J1-DVD.
- [197] Dange, R., Attar, E., Ghodake, P., & Godase, V. (2023). Smart agriculture automation using ESP8266 NodeMCU. *J. Electron. Comput. Netw. Appl. Math.*, (35), 1-9.
- [198] Godase, V. (2025). Optimized Algorithm for Face Recognition using Deepface and Multi-task Cascaded Convolutional Network (MTCNN). *Optimum Science Journal*.
- [199] Mane, V. G. A. L. K., & Gangonda, K. D. S. Pipeline Survey Robot.
- [200] Godase, V. (2025). Navigating the digital battlefield: An in-depth analysis of cyber-attacks and cybercrime. *International Journal of Data Science, Bioinformatics and Cyber Security*, 1(1), 16-27.
- [201] Godase, V., & Jagadale, A. (2019). Three element control using PLC, PID & SCADA interface. *International Journal for Scientific Research & Development*, 7(2), 1105-1109.
- [202] Godase, V. (2025). Edge AI for Smart Surveillance: Real-time Human Activity Recognition on Low-power Devices. *International Journal of AI and Machine Learning Innovations in Electronics and Communication Technology*, 1(1), 29-46.
- [203] Godase, V., Modi, S., Misal, V., & Kulkarni, S. (2025). LoRaEdge-ESP32 synergy: Revolutionizing farm weather data collection with low-power, long-range IoT. *Advance Research in Analog and Digital Communications*, 2(2), 1-11.
- [204] Godase, V. (2025). Comparative study of ladder logic and structured text programming for PLC. Available at SSRN 5383802.
- [205] Godase, V., Modi, S., Misal, V., & Kulkarni, S. Real-time object detection for autonomous drone navigation using YOLOv8. *Advance Research in Communication Engineering and its Innovations*, 2(2), 17-27.
- [206] Godase, V. (2025). Smart energy management in manufacturing plants using PLC and SCADA. *Advance Research in Power Electronics and Devices*, 2(2), 14-24.
- [207] Godase, V. (2025). IoT-MCU Integrated Framework for Field Pond Surveillance and Water Resource Optimization. *International Journal of Emerging IoT Technologies in Smart Electronics and Communication*, 1(1), 9-19.
- [208] Godase, V. (2025). Graphene-Based Nano-Antennas for Terahertz Communication. *International Journal of Digital Electronics and Microprocessor Technology*, 1(2), 1-14.
- [209] Godase, V., Khiste, R., & Palimkar, V. (2025). AI-Optimized Reconfigurable Antennas for 6G Communication Systems. *Journal of RF and Microwave Communication Technologies*, 2(3), 1-12.
- [210] Bhaganagare, S., Chavan, S., Gavali, S., & Godase, V. V. (2025). Voice-Controlled Home Automation with ESP32: A Systematic Review of IoT-Based Solutions. *Journal of Microprocessor and Microcontroller Research*, 2(3), 1-13.
- [211] Jamadade, V. K., Ghodke, M. G., Katakdhond, S. S., & Godase, V. A Comprehensive Review on Scalable Arduino Radar Platform for Real-time Object Detection and Mapping.
- [212] Godase, V. (2025). Cross-Domain Comparative Analysis of Microwave Imaging Systems for Medical Diagnostics and Industrial Testing. *Journal of Microwave Engineering & Technologies*, 12(2), 39-48p.
- [213] V. K. Jamadade, M. G. Ghodke, S. S. Katakdhond, and V. Godase, —A Review on Real-time Substation Feeder Power Line Monitoring and Auditing Systems,” *International Journal of Emerging IoT Technologies in Smart Electronics and Communication*, vol. 1, no. 2, pp. 1-16, Sep. 2025.
- [214] V. V. Godase, “VLSI-Integrated Energy Harvesting Architectures for Battery-Free IoT Edge Systems,” *Journal of Electronics Design and Technology*, vol. 2, no. 3, pp. 1-12, Sep. 2025.
- [215] A. Salunkhe et al., “A Review on Real-Time RFID-Based Smart Attendance Systems for Efficient Record Management,” *Advance Research in Analog and Digital Communications*, vol. 2, no. 2, pp.32-46, Aug. 2025.
- [216] Vaibhav, V. G. (2025). A Neuromorphic-Inspired, Low-Power VLSI Architecture for Edge AI in IoT Sensor Nodes. *Journal of Microelectronics and Solid State Devices*, 12(2), 41-47p.



- [217] Nagane, M.S., Pawar, M.P., & Godase, P.V. (2022). Cinematica Sentiment Analysis. *Journal of Image Processing and Intelligent Remote Sensing*.
- [218] Godase, V.V. (2025). Tools of Research. SSRN Electronic Journal.
- [219] Godase, V. (n.d.). EDUCATION AS EMPOWERMENT: THE KEY TO WOMEN'S SOCIO ECONOMIC DEVELOPMENT. *Women Empowerment and Development*, 174–179.
- [220] Godase, V. (n.d.). COMPREHENSIVE REVIEW ON EXPLAINABLE AI TO ADDRESSES THE BLACK BOX CHALLENGE AND ITS ROLE IN TRUSTWORTHY SYSTEMS. In *Sinhgad College of Engineering, Artificial Intelligence Education and Innovation* (pp. 127–132).
- [221] Godase, V. (n.d.-b). REVOLUTIONIZING HEALTHCARE DELIVERY WITH AI-POWERED DIAGNOSTICS: A COMPREHENSIVE REVIEW. In *SKN Sinhgad College of Engineering, SKN Sinhgad College of Engineering* (pp. 58–61).
- [222] Dhope, V. (2024). SMART PLANT MONITORING SYSTEM. In *International Journal of Creative Research Thoughts (IJCRT)*. <https://www.ijcrt.org>
- [223] M. M. Zade, Sushant D. Kambale, Shweta A. Mane, Prathamesh M. Jadhav. (2025) "IOT Based early fire detection in Jungles". *RIGJA&AR* Volume 2 Issue 1, ISSN:2998-4459. DOI: <https://doi.org/10.5281/zenodo.15056435>
- [224] M. M. Zade, Bramhadev B. Rupanar, Vrushal S. Shilawant, Akansha R. Pawar (2025) "IOT Flood Monitoring & Alerting System using Raspberry Pi-Pico" *International Journal of Research Publication & Reviews*, Volume 6, Issue 3, ISSN:2582-7421. DOI: <https://ijrpr.com/uploads/V6ISSUE3/IJRPR40251.pdf>
- [225] M.M.Zade (2022) "Touchless Fingerprint Recognition System" (Paper-ID 907) (2022) *International Conference on "Advanced Technologies for Societal Applications: Techno-Societal 2022"* <https://link.springer.com/book/10.1007/978-3-031-34644-6?page=6>
- [226] Mr. M.M. Zade published the paper on "Automation of Color Object Sorting Conveyor Belt", in *International Journal of Scientific Research in Engineering & Management (IJSREM)*, ISSN:2582-3930 Volume 06, Issue 11th November 2022.
- [227] Mr. M.M. Zade published the paper on "Cloud Based Patient Health Record Tracking web Development", in *International Journal of Advanced Research in Science, Communication & Technology (IJAR SCT)*, ISSN NO:2581-9429 Volume 02, Issue 03, DOI 1048175/IJAR SCT-3705, IF 6.252, May 2022.
- [228] Mr. Mahesh M Zade, "Performance analysis of PSNR Vs. Impulse Noise for the enhancement of Image using SMF", *Journal of Applied Science & Computations (JASC UGC Approved)*, Volume VI, Issue II, Feb. 2019
- [229] Mr. Mahesh M Zade, "Classification of Power Quality Disturbances Using SVM & their Efficiency Comparison", *Journal of Applied Science & Computations (JASC UGC Approved)*, Volume VI, Issue II, Feb. 2019
- [230] Mr. Mahesh M Zade, "Dynamic Clustering of Wireless Sensor Network Using Modified AODV", *Journal of Applied Science & Computations (JASC UGC Approved)*, Volume VI, Issue II, Feb. 2019
- [231] Mr. Mahesh M Zade, "Performance analysis of PSNR Vs. Impulse Noise for the enhancement of Image using SMF", *National Conference on Mathematical Modeling and Computational Intelligence 2K19 (MMCI-2k19)*, in association with JASC, at S. B. Patil College of Engineering, Indapur, Feb. 2019
- [232] Mr. Mahesh M Zade, "Classification of Power Quality Disturbances Using SVM & their Efficiency Comparison", *National Conference on Mathematical Modeling and Computational Intelligence 2K19 (MMCI-2k19)*, in association with JASC, at S. B. Patil College of Engineering, Indapur Feb. 2019
- [233] Mr. Mahesh M Zade, "Dynamic Clustering of Wireless Sensor Network Using Modified AODV", *National Conference on Mathematical Modeling and Computational Intelligence 2K19 (MMCI-2k19)*, in association with JASC, at S. B. Patil College of Engineering, Indapur Feb. 2019
- [234] Mr. Mahesh M Zade & Mr. S.M. Karve, "Performance Analysis of Median Filter for Enhancement of Highly Corrupted Images", *National Conference on Advanced Trends in Engineering*, Association with IRJMS, Karmyogi Engineering College, Shelave, Pandharpur, March 2016.



- [235] Mr. Mahesh M Zade & Mr.S.M.Karve,"Implementation of Reed Solomen Encoder & Decoder Using FPGA",National Conference on Advanced Trends in Engineering, Association with IRJMS, Karmyogi Engineering College, Shelave, Pandharpur, March 2016.
- [236] Mr. Mahesh M Zade & Dr.S.M.Mukane,"Performance of Switching Median Filter for Enhancement of Image", National Conference on Mechatronics at Sinhgad Institute of Technology and Science, Narhe, Pune, Feb. 2016.
- [237] Mr. Mahesh M Zade & Dr.S.M.Mukane,"Enhancement of Image with the help of Switching Median Filter", National Conference on Emerging Trends in Electronics & Telecommunication Engineering, SVERI's College of Engineering Pandharpur, NCET 2013.
- [238] Mr.Mahesh M Zade & Dr.S.M.Mukane,"Enhancement of Image with the help of Switching Median Filter", International Journal of Computer Application (IJCA) SVERI's College of Engineering, Pandharpur, Dec.2013.
- [239] A. O. Mulani, V. Godase, S. Takale, and R. Ghodake, "Secure Image Authentication using AES and DWT Watermarking on Reconfigurable Platform," International Journal of Embedded System and VLSI Design, vol. 1, no. 2, pp. 14-20, Oct. 2025.

